# Amal Mehta

SF Bay Area · amal.mehta@gmail.com · amalmehta.com · LinkedIn · GitHub

Research engineer with six years of applied deep learning, from generative models and vision to deep reinforcement learning. Earlier neuroimaging research at UCSF, and a long-standing interest in how the brain learns.

## Experience

**Independent ML & Robotics Research** **Sept 2025 – Present**
*Self-directed projects; details at amalmehta.com* *San Francisco Bay Area*

- **Reward hacking in code agents:** GRPO on visible tests taught an agent to delete 98% of untested branches (true correctness 31% → 2% as pass rate rose); random 25% hidden-check audits stopped it in the pilot, but 50 follow-ups found they only delay it.
- **Brain–computer interfaces:** four projects on public neural data; label-free decoding averaged $R^2$ 0.72 on 17 of 19 held-out days to day 138 (Wiener 0.42).
- **Post-training:** SFT then GRPO on a small transformer policy in a key-and-door gridworld maze; over 3 seeds, GRPO lifted held-out success from 70% to 80% (DPO: 81%), where RL from scratch failed.
- **Meta-RL and self-play:** quadrotor adapting to payload, wind and motor wear with seven meta-learning methods; 9×9 Go by self-play.
- **Tools:** LLM Understander, an interactive course on how LLMs are trained; Researcher Pal, Claude Code skills for research papers.
- Mentored undergraduate scholars from Berkeley.

**Deep Learning Scientist** **Sept 2021 – Aug 2025**
*PhysicsAI (DARPA-funded)* *Pacifica, CA (remote)*

- Formulated synthetic-data generation as RL: policies (TD3, PPO, DDPG) set image-generator parameters, rewarded by detector performance on real drone imagery, without hand-tuned rendering.
- Built the PEARL meta-RL codebase (task embeddings over TD3) and ran its multi-week GPU training, for few-shot generator adaptation.
- Implemented domain adaptation from scratch to hold detector accuracy with scarce real labels, on 10–15 GB mostly synthetic datasets.
- Built GAN-based synthesis pipelines (DCGAN, InfoGAN, FineGAN) producing training data for downstream detection models.

**Deep RL Researcher** **Aug 2023 – Mar 2024**
*Hybrid Robotics Lab, UC Berkeley — adv. Prof. Koushil Sreenath* *Berkeley, CA*

- Trained a single multi-stage PPO policy in IsaacGym for a Unitree Go1 quadruped to track, approach, and shoot a soccer ball at a target, replacing the multi-policy hierarchical framework used in prior work; ∼60% kicking accuracy on hardware.
- Designed the three-stage reward and transition machinery and CPG-based gait tracking; prototyped RLPD (SAC with prior data) to fine-tune the sim-trained policy on real robot data.

**Research Assistant** **June 2020 – Dec 2022**
*Video & Image Processing Lab, BAIR — adv. Prof. Avideh Zakhor* *UC Berkeley*

- Developed a deep CNN segmentation pipeline for invasive melanoma in large-scale histopathological images (F1 0.44 vs. 0.14 for the prior best); published at SPIE Medical Imaging 2023; Stowell-Orbison Award Poster Session at USCAP 2022.
- Segmented the epidermis and melanoma separately at four magnifications, on slides from 57 UCSF patients.

**Researcher** **June 2020 – Aug 2020**
*Abbasi-Asl Lab, UCSF — adv. Prof. Reza Abbasi-Asl* *San Francisco, CA*

- Built empirical mode decomposition (EMD) pipelines to align 2D and 3D brain scans across visits and to a common template.

## Education

**University of California, Berkeley** **Aug 2017 – Aug 2021**
*B.A. Computer Science — Regents Scholar* *Berkeley, CA*

- Focused on robotics and linear system theory; Regents, merit, and leadership awards.
- CS285 (Deep RL) project: iterative retraining of PPO agents against adversarial policies.

**Stanford University** **Aug 2021 – May 2023**
*Graduate coursework — Autonomous Systems & Artificial Intelligence*

- CS231N (Deep Learning for Vision), CS330 (Meta-Learning), AA271A/B (Autonomy I & II), AA273 (State Estimation).
- CS330 project: few-shot imitation learning; implemented $RL^2$ and inverse soft Q-learning in Meta-World.

## Technical Skills

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| **Languages** | Python, C++, Swift, JavaScript / TypeScript |
| **ML / RL** | PyTorch, TensorFlow, stable-baselines3, IsaacGym, OpenCV, CTC decoding |
| **Product** | SwiftUI, Three.js |
| **Robotics** | ROS, Gazebo, Unitree Go1, CPG gait control, Kalman filtering, Shared autonomy |
| **Tools** | Docker, GitHub Actions CI, CUDA/multi-GPU |

## Publications

1. Shah, A., **Mehta, A.**, Wang, M., Neumann, N., McCalmont, T., Zakhor, A. "Deep Learning Segmentation for Invasive Melanoma." *SPIE Medical Imaging*, 2023.
2. Neumann, N., Wang, M., **Mehta, A.**, et al. "Quantifying Invasive Melanoma Volume by Deep Learning Segmentation at the Pixel-Level." *USCAP Annual Meeting*, 2022.